

| Question | Marking Guidance                                                                                                                                                                                                                                                                                   | Mark | Comments                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------|
| 08.1     | Calculations made (from raw data)/raw data would have recorded initial and final masses;                                                                                                                                                                                                           | 1    |                                                                               |
| 08.2     | Add 4.5 cm <sup>3</sup> of (1.0 mol dm <sup>-3</sup> ) solution to 25.5 cm <sup>3</sup> (distilled) water;                                                                                                                                                                                         | 2    | If incorrect, allow 1 mark for solution to water in a proportion of 0.15:0.85 |
| 08.3     | 1. Water potential of solution is less than/more negative than that of potato tissue;<br>2. Tissue loses water by osmosis;                                                                                                                                                                         | 2    | 1. Allow $\Psi$ as equivalent to water potential                              |
| 08.4     | 1. Plot a graph with concentration on the x-axis and percentage change in mass on the y-axis;<br>2. Find concentration where curve crosses the x-axis/where percentage change is zero;<br>3. Use (another) resource to find water potential of sucrose concentration (where curve crosses x-axis); | 3    |                                                                               |